Kaspa Network Congestion Vulnerability

Analysis of Testnet-10 Storm Results, Economic Attack Costs, Threats to Sequencing Infrastructure, and Mitigation Options

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1. Introduction

This document summarises empirical testing conducted on Kaspa Testnet-10 (TN10) between September and October 2026, together with an economic assessment of equivalent activity on mainnet. The testing demonstrated that sustained transaction submission rates exceeding the network's practical simple-payment capacity (approximately 2,500–3,400 transactions per second at 10 blocks per second) are achievable with modest infrastructure. Excess transactions enter the mempool, producing measurable delays for lower-priority traffic. Because Kaspa's design objective includes support for high-throughput sequencing of critical infrastructure and application data, the existence of an economically accessible method to occupy block mass constitutes a material operational risk. This report presents the evidence, quantifies costs under current and hypothetical token prices, evaluates secondary market incentives, and outlines mitigation measures that can be implemented by application developers and at the protocol level.

2. Conclusion

Testnet results establish that the Kaspa network can be driven into a congested state by coordinated simple-payment submission at rates above approximately 3,000 transactions per second. The pure fee cost of sustaining such activity for intervals of 5–120 minutes is in the low-to-mid tens of thousands of euros at a token price of €1, and scales linearly with price and duration. An attacker who simultaneously holds a short position can, under certain market conditions, offset or exceed this cost through a decline in the token price. The vulnerability is therefore both technical and economic. It is not closed by current protocol parameters. Critical applications that sequence data or instructions on Kaspa must implement network monitoring and dynamic fee escalation. Protocol-level increases in block rate and throughput, already listed on the development roadmap, raise the absolute cost of saturation and should be prioritised. Until these measures are in place, the residual capacity for deliberate congestion remains a backdoor that can affect the reliability of infrastructure dependent on timely inclusion.

3. Background and Empirical Evidence

All quantitative results referenced below are drawn from the public repository **STP-KAS/tn-master** and the individual storm repositories it indexes, principally **tn10-storm-2026-10-10** and **tn10-storm-2026-10-11**. Independent validation of accepted-in-block counts was performed by Kaspa Pulse using two external nodes. No monetary value was involved; all activity used testnet tokens.

3.1 Principal Storm Results

- **10–11 October 2026 storm**: Active window approximately 8 hours. Peak median step ~2,642 tx/s. Network-unique accepted transactions ~32.3 million. Fee rates held at 400/600 sompi per gram.
- **11 October 2026 storm**: Duration 303 minutes (07:30–12:33 UTC). Mean ~2,833–2,844 tx/s; median ~3,107 tx/s; peak complete minute 3,583 tx/s (Kaspa Pulse 3,646 tx/s). Total unique accepted transactions ~51.2–51.5 million. Fee rates began at 400/600 sompi/gram and were raised to 560/840 and later 1,120/1,680 sompi/gram in response to rising activity.

These runs confirm that rates above the commonly cited simple-payment capacity of the 10 BPS network are reachable. Transactions that cannot be included immediately accumulate in the mempool, delaying lower-fee traffic. The same behaviour is expected on mainnet once organic plus adversarial load exceeds block mass limits (500,000 grams per block across compute and storage dimensions).

4. Technical Characteristics of the Vulnerability

Kaspa produces 10 blocks per second. Each block is limited to 500,000 grams of mass in the compute and storage dimensions (transient mass limit is higher). An ordinary one-input, two-output payment consumes approximately 1,600–2,000 grams. The resulting theoretical capacity lies in the range of roughly 2,500–3,400 simple transactions per second. When submission rates exceed inclusion capacity, the mempool grows. Inclusion of any given transaction then depends on its fee rate relative to competing transactions. Critical sequencing traffic that does not pay a competitive fee rate experiences increased latency or temporary exclusion. Because the protocol does not currently distinguish sequencing or covenant traffic from ordinary payments for prioritisation purposes beyond the fee market, any sufficiently resourced actor can occupy block space by submitting simple payments.

5. Economic Cost of Sustained Congestion

Costs below are calculated for a sustained rate of 3,000 transactions per second, a mass of 2,000 grams per transaction, and a fee rate of 400 sompi/gram (normal) or 520 sompi/gram (+30 % priority). One KAS equals 100,000,000 sompi. These rates match the lower end of the fee levels used successfully in the October storms. Higher rates observed during the 11 October run would increase costs further. Infrastructure and UTXO-preparation costs are excluded; only protocol fees are shown.

5.1 Cost at €1.00 per KAS

Duration	Transactions	Normal fee (€)	+30% priority (€)
5 min	900,000	7,200	9,360
10 min	1,800,000	14,400	18,720
15 min	2,700,000	21,600	28,080
30 min	5,400,000	43,200	56,160
1 hour	10,800,000	86,400	112,320
2 hours	21,600,000	172,800	224,640

At a token price of €2 the figures double. At the prevailing market price of approximately €0.038 the absolute cost is correspondingly lower, but the same linear relationship holds. A two-hour attack at 3,000 tx/s therefore requires on the order of €170,000–€225,000 in fees at €1 per KAS, or roughly half that amount at current prices. These sums are within the reach of well-capitalised actors.

6. Threat Assessment

6.1 Direct Operational Impact

Any application that relies on timely inclusion of sequencing transactions, data commitments, or agent instructions is exposed. During periods of mempool growth, transactions that do not outbid the congestion traffic experience delayed or failed inclusion. For infrastructure that assumes sub-second or low-second

finality, intervals of 5–20 minutes of elevated latency are operationally material. The testnet storms already demonstrated measurable disruption to concurrent activity.

6.2 Market and Incentive Effects

An attacker who opens or holds a short position in KAS can treat the fee expenditure as a cost of generating negative sentiment. A publicly visible congestion event lasting even a few tens of minutes can prompt selling, particularly on a market whose liquidity is limited. If the resulting price decline exceeds the fee outlay (after trading costs and slippage), the attack is profitable on a net basis. The combination of a documented, low-complexity method and an available shorting market creates a concrete economic incentive. The existence of this incentive does not require the attacker to possess a large fraction of hashrate; fee-paying submission alone is sufficient.

It is important to note that a single short window does not automatically produce a lasting price collapse. Markets frequently treat isolated stress events as temporary. Repeated or prolonged episodes, or clear failures of high-profile applications, increase the probability of a larger and more persistent reaction. The residual risk remains material precisely because the cost of producing the initial visible disruption is not prohibitive.

7. Mitigation Options

7.1 Application-Level Measures (Immediate)

Critical infrastructure that sequences on Kaspa should implement continuous monitoring of mempool size, block fill ratios, and own-transaction inclusion latency. Upon detection of rising congestion, the application must automatically elevate its fee rate—initially to the priority bucket, and further if required—to maintain inclusion priority. This forces any attacker who wishes to continue outbidding the protected traffic to raise its own fee rate, producing a progressive increase in attack cost. The mechanism is already supported by the node’s fee-estimate RPC and requires no protocol change. Developers building on Kaspa should treat dynamic fee escalation as a mandatory component of production deployments.

7.2 Protocol-Level Measures

Increasing the block rate and/or the mass limit per block raises the absolute number of transactions that must be submitted to reach saturation. Both increases are already present on the Kaspa development roadmap. Higher throughput directly elevates the fee cost of an attack of any given duration and reduces the window during which ordinary traffic is displaced. Additional protocol options that merit evaluation include differentiated treatment of sequencing or covenant transactions in the mempool selection algorithm, and the ability for applications to signal priority class without relying solely on fee rate.

7.3 Supplementary Measures

• Application designs that can tolerate bounded delays (for example, by batching or by accepting slightly later inclusion without breaking invariants) reduce the practical impact of short congestion windows.
• Public transparency of mempool and fee-market conditions allows independent observers to detect anomalous submission patterns early.
• Continued public stress testing on testnet, under controlled conditions, improves the accuracy of capacity models and the calibration of monitoring thresholds.

8. Recommendations

1. Application developers deploying sequencing or critical infrastructure on Kaspa should implement network monitoring and automatic fee escalation as a standard production requirement before mainnet reliance.

2. Protocol development should continue to prioritise increases in block rate and throughput, as these measures raise the economic cost of saturation attacks in a manner that benefits all users.

3. The residual capacity for deliberate congestion should be treated as an open issue until the combination of higher throughput and widespread adoption of dynamic fee management materially reduces the feasibility of short-duration disruption.

4. Documentation and reference implementations of congestion-aware fee management should be made available to the developer community so that the mitigation is not left to ad-hoc implementation.

Sources: STP-KAS/tn-master and linked storm repositories (October 2026); Kaspa network parameters (10 BPS, 500,000 g block mass, 100 sompi/gram minimum relay post-Toccata); independent accepted-in-block counts by Kaspa Pulse. All cost figures are derived from the stated assumptions and the empirical fee rates used in the referenced storms.